INCH-POUND

MIL-C-22244D(AS)
30 September 1994
SUPERSEDING
MIL-C-22244C(AS)
10 December 1985

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 23 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Departments of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the Mark 23 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issuer of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-W-83420	Wire Rope, Flexible, for Aircraft Control

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Systems Standardization, Code 4.1.11B120-3, Naval Air Warfare Center Aircraft Division, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk,700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (30003)

DL 419687	Container, Inner, for Cartridge, Impulse, Mark 23 Type
DL 419688	Cartridge, Outer, for Cartridge, Impulse, Mark 23 Type
DL 491837	Cartridge, Impulse, Mark 23 Mod 0
1861735	Unloaded Shell Assembly
2022774	Cartridge, Impulse, Mark 23 Mod 0 Assembly

(Copies are available from the Commander, Naval Surface Warfare Center, Data Control Branch (Code 8410), Indian Head Division, Indian Head, MD 20640-5035.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OP 5	Ammunition and Explosives Ashore

(Copies are available from the Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199	Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 545 Image Quality in Direct Thermal Neutron Radiographic Examination

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Conformance to documents. The Mark 23 Mod 0 Impulse Cartridge shall be manufactured in accordance with DL 491837.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Bridgewire. The bridgewire shall be soldered in place as specified on Drawing 1861735.

3.5 Bridgewire continuity. The bridgewire shall glow when subjected to the continuity test (see 4.6.4).

3.6 Unloaded shell assembly leakage. The unloaded shell assembly shall not leak (see 4.6.5).

3.7 Propellant charge.

3.7.1 Quantity. All propellant charges shall be as specified on Drawing 2022774. Charges shall be determined by weight rather than by volume measurement.

3.7.2 Smokeless powder, handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.7.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.6.1).

3.8 Cartridge inspection.

3.8.1 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.6.3), shall be not less than 3.0 ohms nor greater than 6.5 ohms.

3.8.2 N-ray. Cartridges shall be free of imperfections when N-rayed (see 4.6.6).

3.9 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.9.1 through 3.9.3 when subjected to the test prescribed in 4.6.7.

3.9.1 Output. Cartridges shall furnish sufficient power to completely sever the steel cable specified in 4.6.7.

3.9.2 Misfire. There shall be no misfires (see 4.6.7.2).

3.9.3 Hangfire. There shall be no hangfires (see 4.6.7.3).

3.10 Workmanship. Cartridges shall be free of the following defects: burrs, dents, missing or damaged O-ring, perforated closure disc, or other defects which may prevent entry of the cartridge into the firing chamber, or affect resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the applicable test method of 4.6.

4.4 First article inspection. First article inspection shall be performed after award of contract and prior to production at a location acceptable to the Government (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.2 through 3.8.2 and 3.10 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, and procedures (see 6.3) normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Unless otherwise specified (see 6.2 and 6.4), 76 cartridges shall be submitted for first article inspection.

4.4.2 Inspection routine. Sixty-six cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests (see 6.3) as specified in MIL-D-21625F (see 6.2b): 40-foot drop test (6); THC (12), except that 14 days shall be used in lieu of 28 days; vibration (12); –65°F temperature (12); 160°F temperature (12); 70°F temperature (12). The number in parenthesis denotes the number of cartridges allocated for each test. No cartridge shall produce results which fail to meet the requirements of 3.9.1 through 3.9.3. Firings shall be conducted in the test fixture of 4.6.7.

4.4.3 Failure. The failure of any cartridge to meet the requirements stated above shall cause rejection of the first article sample.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests (see 6.3) specified in table I and verification of the characteristics classified on the drawings of DL 491837.

4.5.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 281 and not greater than 10,000. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

4.5.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II or MIL-STD-105.

TABLE I. Quality conformance tests and examinations.

Item	Nature of Test	Requirement paragraph	Test paragraph	Classification
Bridgewire	Soldering	3.4	—	(M101)
Bridgewire	Continuity	3.5	4.6.4	(C1)
Unloaded shell assembly	Leakage	3.6	4.6.5	(C2)
Propellant charge	Type and weight	3.7.1	—	(C3)
Smokeless powder	Condition	3.7.3	4.6.1	(C4)
Cartridge	Visual inspection	3.8.3	4.6.2	(M102)
Cartridge	Bridgewire resistance	3.8.1	4.6.3	(C5)
Cartridge	N-ray	3.8.2	4.6.6	(C6)
Cartridge	Ballistics	3.9	4.6.7	(C7)
Cartridge	Preservation-packaging, packing, and marking	Section 5	4.6.2	(M103)

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
51 to 90	13	10
91 to 150	20	10
151 to 280	32	10
281 to 500	50	10
501 to 1200	80	10
1201 to 3200	125	10
3201 to 10,000	200	10
10,001 to 35,000	315	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Tests.

4.6.1 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.6.2 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.8.3 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5.

4.6.3 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.8.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to not greater than 50 milliamperes. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 1 percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.8.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.6.4 Bridgewire circuit continuity. After soldering the bridgewire, the unloaded shell assembly (Drawing 1861735) shall be thoroughly cleaned to remove flux and contamination. After cleaning, the bridgewire shall be flashed with an electric current using a voltage source of 6 volts ac or dc to produce a visible red glow of the bridge and ensure continuity. A suitable resistor may be used to regulate the current to the bridge. Flashing is a necessary manufacturing operation and shall be conducted on each loaded shell assembly. Assemblies with bridgewires which fail to glow shall be cause for rejection. Any faulty assemblies may be reworked by replacement of the bridges.

4.6.5 Unloaded shell assembly leakage. Apply not less than 10 psi of air pressure internally to all unloaded shell assemblies (Drawing 1861735) which have passed the continuity test of 4.6.4 and check for leaks. All assemblies that show evidence of leakage shall be cause for rejection.

4.6.6 Radiographic inspection. All cartridges shall be radiographically examined in accordance with ASTM E 545. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the N-ray. Defective cartridges found by the vendor shall be clearly identified on the N-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.6.7 Ballistic test. The number of cartridges from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned not less than 6 hours at 70 ± 5°F and fired in a cable cutter, Mine Safety Appliance Company Part No. TA 97116 or Part No. TA 97999, or equal. A 3/16-inch-diameter steel cable in accordance with MIL-W-83420, Type I, Composition B, shall be properly positioned in the cable cutter for cutting. The cable cutter shall be adapted to measure the time from actuation of the firing mechanism to complete severance of the cable. The firing current shall be 4.0 ± 0.5, − 0 amperes. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes.

4.6.7.1 Criteria of acceptability. The failure of any cartridge to completely sever the steel cable shall be cause for rejection of the lot represented.

4.6.7.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.6.7.3 Hangfire. One or more hangfires (see 6.5.5) shall be cause for rejection of the lot represented.

4.7 Retest. There shall be no retests.

4.7.1 Test failure. If test failure is attributable to an assignable cause, excluding test cartridge, the original test results should be discarded and that portion of the test reconducted.

5. PACKAGING

5.1 Preservation and packaging. For definitions of packaging levels see 6.5.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with DL 419687.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed in the contract (see 6.2), those cartridges intended for expenditure in first article sample testing and for production lot

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

acceptance testing shall be packaged in accordance with Method III of MIL-P-116 and with 49 CFR 171-178.

5.2 Packing. For definitions of packing levels see 6.5.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with DL 419688.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When specified in the contract (see 6.2), cartridges intended for expenditure in first article sample testing and for production lot acceptance testing are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with 49 CFR 171-178.

5.3 Marking.

5.3.1 Special marking. Marking of exterior containers shall be in accordance with 49 CFR 171-178.

5.3.2 Normal marking. Unless otherwise specified in the contract or order (see 6.2), the marking information on unit packages and shipping containers shall be as specified on the drawings of DL 419687 and DL 419688, respectively.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Mark 23 Mod 0 Impulse Cartridge is intended to actuate a helicopter-rescue hoist cable cutter (MSA Part No. TA 97116 or Part No. TA 97999, or equal) to jettison a load in an emergency.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required and if so specify the test activity (see 3.1 and 4.4).

d. Date and place of delivery.

e. Inspection conditions, if other than as specified in 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

f. Sample size, if other than as specified in 4.4.1 and 4.5.2.

g. Level C packaging, if other than as specified (see 5.1.3 and 5.2.3).

h. Marking, if other than as specified in 5.3.2.

i. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity the safety precaution requirements of "Ammunition and Explosives," OP 5, are applicable.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractural application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data. except where DOD FAR Supplement 227.405-70 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4, 4.5	DI-NDTI-80808	Test plans/procedures	---
4.4.2, 4.5	DI-NDTI-80809A	Test inspection reports	---

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 76 cartridges. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of the of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions. For the purpose of this specification the following

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.

definitions apply.

6.5.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical handling during shipment, indeterminate storage and world-wide redistribution.

6.5.2 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.5.3 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide distribution.

6.5.4 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to the applicable carrier rules and regulations and may be the supplies's commercial practice when such meets the requirement of this level.

6.5.5 Hangfire. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and complete severance of the steel cable exceeds 0.250 second.

6.6 Subject term (key word) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review activity:
Navy - OS

Preparing activity:
Navy - AS

(Project 1377-NE67)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:32Z
Check the source to verify that this is the current version before use.